

PAPER_202: UQFF Reionization, BBN, Recombination, and Cosmic Dawn Physics

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

Abstract

Key UQFF calibrated constants: $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$; $[SSq] = 5.7\text{e-}1$; $H_{SCm} \approx 9.9\text{e-}1$; $U_{UA} \approx 1.0\text{e-}4$; $k_{\eta} = 1.0\text{e-}113$; $\beta_i \approx 6.0\text{e-}1$; $G = 6.674\text{e-}11 \text{ N}\cdot\text{m}^2/\text{kg}^2$

This paper presents the UQFF buoyancy formalism applied to cosmic dawn and reionization physics: baryon-to-photon ratio (?), deuterium bottleneck during Big Bang Nucleosynthesis (BBN), CMB angular power spectrum, recombination optical depth, ionization fraction evolution, HII bubble growth rate, and Jeans mass/wavelength for fragmentation. These collectively span the redshift range $z \sim 1100$ (recombination) through $z \sim 5$ (end of reionization). Each F_{UBii} and U_m variant is rigorously derived from observational anchor equations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Baryon-to-Photon Ratio (?)

$\eta = n_b/n_\gamma = 6.08 \times 10^{-10}$ (Planck 2018 + BBN fit)

$F_{UBii, \eta} = F_{rel} \times (\eta = n_b/n_\gamma / E_{LEP}) \times Q_{wave} \times [\text{Neff modification}]$

$U_{m, \eta}(t) = \mu(\eta_{vac}) \cdot (1 - e^{-\eta t}) \cdot \eta$

Physical origin:

$n_\gamma = 410 \text{ cm}^{-3}$ (CMB photon number density today)

Fit anchor: Primordial D, ^3He , ^4He , ^7Li abundances

η determines all light element yields (standard BBN)

UQFF role: η sets vacuum energy scale through η_b/η_γ at nucleosynthesis

2. Deuterium Bottleneck (BBN Timing)

Deuterium bottleneck timescale:

$$t_D \sim v(3/(32pG\gamma_{\text{rad}})) \sim 180 \text{ s at } T \sim 0.1 \text{ MeV}$$

$$\gamma_{\text{rad}} = p^2 k T^4 / (30 h^3 c^5) \cdot g_*$$
 (effective DOF $g_* \sim 10$ at D-formation onset)

Sequence:

$T \sim 1 \text{ MeV}$: neutron freeze-out ($n/p \sim 1/7$)

$T \sim 0.1 \text{ MeV}$ ($t \sim 180 \text{ s}$): D photodissociation ends ? 4He chains proceed

Yield: $Y_P \sim 0.247$ (mass fraction 4He)

$$F_{\text{UBii,deb}} = -F_{\text{rel}} \times (t_D / E_{\text{LEP}}) \times Q_{\text{wave}} \times [g_*(T) \text{ variation}]$$

$$U_{\text{m,deb}}(T) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot [\text{Weak freeze at } T \sim 1 \text{ MeV; D photodissociation below}]$$

UQFF calibration: ? calibrated to UQFF k_* parameter through BBN freeze-out

$k_* \sim 10^{11.3}$ (small-scale vacuum fluctuation coupling)

γk_* (CIA cross-section refit) $\sim 7.25 \times 10^8$ (fractional shift from H2 CIA data)

3. CMB Angular Power Spectrum

$$C_l = (2/p) \cdot \gamma k^2 dk \cdot P(k) \cdot |\gamma_l^T(k)|^2$$

$P(k) \propto k^{n_s-4}$ (primordial, $n_s \sim 0.965$ Planck 2018)

$\gamma_l^T(k)$ = transfer function:

- Large scales ($l < 100$): Sachs-Wolfe plateau $C_l \sim \text{constant}$
- Acoustic peaks ($l = 220, 540, 800 \dots$): baryon-photon oscillations
- Damping tail ($l > 1000$): Silk damping

$$F_{\text{UBii,cmb}} = F_{\text{rel}} \times (C_l \cdot l(l+1)/(2p) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,cmb}}(k) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot \text{Transfer } \gamma_l^T: \text{Sachs-Wolfe + acoustic oscillations}$$

Calibration anchor:

$D_l = l(l+1)C_l/(2p) \times T_0^2$? peak at $l \sim 220$ matches $0_{\text{tot}} = 1$

UQFF role: vacuum energy $\gamma c^2/3$ term in g_{UQFF} sets acoustic horizon

4. Recombination Optical Depth

$$t(z) = \gamma_z^8 n_e(z') \cdot s_T \cdot c \cdot |dt/dz'| dz'$$

$s_T = 6.652 \times 10^{-29} \text{ m}^2$ (Thomson cross-section)

$$n_e(z) = x_e(z) \cdot n_H(z) = x_e(z) \cdot n_{b,0} \cdot (1+z)^3$$

Recombination: $z_{\text{rec}} \sim 1100$? $t(z_{\text{rec}}) \sim 1$ (photon decoupling)

Reionization: $z_{\text{re}} \sim 7.7$? optical depth $t_{\text{re}} \sim 0.054$ (Planck 2018)

$$F_{\text{UBii,recomb}} = -F_{\text{rel}} \times (t(z) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,recomb}}(z) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot n_e \cdot s_T \cdot c \cdot dt$$

Physical context:

After z_{rec} , photons decouple and stream freely (CMB)

Surface of last scattering has $T/T \sim 10^5$ (measured by Planck)

5. Reionization: Ionization Fraction Evolution

$$dx_e/dt = \Gamma_{\text{eff}} \cdot e_{\text{esc}} \cdot f_{\text{eff}} - a_B \cdot n_e^2 \cdot C$$

where:

$\Gamma_{\text{eff}} = (1+z)^3 \cdot n_b \cdot \Gamma_{\text{eff}}$ (ionizing photon rate)

$e_{\text{esc}} \sim 0.1-0.3$ (photon escape fraction from galaxies)

$f_{\text{eff}} \sim 0.05-0.2$ (star formation efficiency in halos)

$a_B = 2.6 \times 10^{-13} \text{ cm}^3/\text{s}$ (recombination coefficient, case B, $T \sim 10^4 \text{ K}$)

$C = n_H^2 / n_H^2$ (clumping factor, $C \sim 3$)

$$F_{\text{UBii,ion}} = F_{\text{rel}} \times (\Gamma_{\text{eff}} \cdot e_{\text{esc}} \cdot f_{\text{eff}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{recombination subtracted}]$$

$$U_{\text{m,ion}}(t) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot a_B \cdot n_e^2 \cdot C \, dt$$

Reionization history:

$z \sim 20-30$: first stars ionize H around them (PopIII)

$z \sim 7-9$: bulk reionization, x_e rises from ~ 0.1 to ~ 1

$z \sim 5-6$: completion (Gunn-Peterson trough in quasar spectra)

6. HII Bubble Growth During Reionization

$$dN_b/dt = \Gamma_{\text{eff}} \cdot e_{\text{esc}} - a_B \cdot n_H \cdot (4/3 \pi R_b^3)$$

Simplified overlap model (Stromgren sphere analog):

$$R_b(t) = (3 \Gamma_{\text{eff}} \cdot t / (4 \pi \cdot n_H))^{1/3} \quad (\text{linear ionization front growth})$$

Γ_{eff} = rate of ionizing photons emitted (from galaxy SFR)

n_H = neutral H density (z -dependent)

$$F_{\text{UBii,bub}} = F_{\text{rel}} \times (R_b(t) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (\Gamma_{\text{eff}} \cdot x_e)$$

$$U_{\text{m,bub}}(t) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [\text{Expand for moving ionization front}]$$

Physical context:

Bubble merger & percolation at $x_{\text{HI}} < 0.1$? full reionization

UQFF buoyancy at bubble edge acts as expansion driver ($F_{\text{UBii,bub}} > 0$)

7. Jeans Length and Mass (Fragmentation)

Jeans length:

$$\lambda_J = p^{1/2} \cdot c_s / (G \cdot \rho)^{1/2}$$

Jeans mass:

$$M_J = (p/6) \cdot \rho \cdot \lambda_J^3 = (5k_{BT}/(G\mu_H))^{\{3/2\}} \cdot (3/(4p))^{\{1/2\}}$$

Stability condition: $\rho > \rho_J$? gravitational collapse

Dispersion relation: $\omega^2 = c_s^2 \cdot k^2 - 4\pi G \rho$

Unstable: $\omega^2 < 0$? $k < k_J = 2\pi/\lambda_J$

$F_{UBii,jeans} = -F_{rel} \times (\lambda_J / E_{LEP}) \times Q_{wave} \times (\text{collapse onset time } t \sim 1/v(G))$

$U_{m,jeans}(T) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot [\text{Perturb: } \omega^2 = c_s^2 \cdot k^2 - 4\pi G \rho]$

Physical context:

First (PopIII) stars: $T \sim 200$ K, cloud $M \sim 100\text{--}1000 M_\odot \gg M_J$

Present-day GMCs: $T \sim 10\text{--}30$ K ? $M_J \sim 1\text{--}10 M_\odot$

UQFF buoyancy at Jeans scale: $F_{UBii,jeans}$ acts as tidal disruption force

8. Alfvén Wave and Turbulent Cascade

Alfvén wave velocity:

$$v_A = B/v(4\pi\rho) \quad (\text{for } B \text{ in Gauss, } \rho \text{ in g/cm}^3)$$

Anisotropic cascade (Goldreich-Sridhar):

$$k_\parallel/k_\perp \sim (k_\perp \cdot l_A)^{\{1/3\}} \quad (\text{scale-dependent anisotropy})$$

$F_{UBii,alf} = F_{rel} \times (v_A / E_{LEP}) \times Q_{wave} \times (B \cdot \rho v_A) \times e^{-t/t_{eddy}}$

Turbulent energy cascade (Kolmogorov in ISM):

$$e = v_l^3/l = \text{constant} \quad (\text{energy transfer rate per unit mass})$$

$$v_l \propto l^{\{1/3\}} \quad (\text{velocity-scale relation})$$

Power spectrum: $E(k) \propto k^{-5/3}$

$F_{UBii,turb} = F_{rel} \times (e^{\{2/3\}} \cdot l^{-\{2/3\}} / E_{LEP}) \times Q_{wave} \times e^{-kl/k_J}$

$$U_{m,alf}(B) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot v_A(B)$$

$$U_{m,turb}(l) = \mu(\rho_{vac}) \cdot (1 - e^{-\rho t}) \cdot (\rho \cdot v_l^3/l)$$

9. Ionization Parameter and Feedback Coupling

Ionization parameter:

$$U = Q_H/(4\pi \cdot r^2 \cdot n_H \cdot c)$$

where:

Q_H = number of ionizing photons per second (from AGN or OB stars)

r = distance from ionizing source

$\log U \sim -2$ to -3 for NLR, -1 to 0 for BLR

AGN feedback coupling efficiency:

$$e_f = E_{kin}/(\rho_{acc} \cdot c^2) \sim 0.05\text{--}0.1$$

$$E_{kin} = (1/2)\rho_{out} \cdot v_{out}^2$$

Only $e_f \times 10^{-5}$ per $M_\odot$ needed to match M_{BH-s}

$$F_{UBii,upar} = F_{rel} \times (U \cdot n_H \cdot c / E_{LEP}) \times Q_{wave} \times (Q_H/r^2)$$

$$F_{UBii,coup} = F_{rel} \times (e_f \cdot \tau_{acc} \cdot c^2 / E_{LEP}) \times Q_{wave} \times [0.05-0.1]$$

$$Um,upar(r) = \mu(\tau_{vac}) \cdot (1 - e^{-\tau t}) \cdot (Q_H/(4\pi r^2 n_H c))$$

$$Um,coup(\tau) = \mu(\tau_{vac}) \cdot (1 - e^{-\tau t}) \cdot e_f \text{ (couple fraction to kinetic energy)}$$

10. Cosmological Timeline Summary

Epoch	Redshift	UQFF Process	F_UBii Variant
Inflation	$z \sim 10^{27}$	Inflaton field V(?)	F_UBii,inf
Planck epoch	$z \sim 10^{32}$	LQC bounce	F_UBii,bounc
Neutron freeze-out	$z \sim 10^{10}$ (T~1 MeV)	Weak interactions	F_UBii,eta
BBN	$z \sim 10^9$ (t~180 s)	D bottleneck	F_UBii,deb
Recombination	$z \sim 1100$	Photon decoupling	F_UBii,recomb
Cosmic dawn	$z \sim 20-30$	First stars, first HII	F_UBii,ion
Reionization	$z \sim 6-9$	HII bubble percolation	F_UBii,bub
Present	$z \sim 0$	Structure + feedback	All F_UBii

11. References

- grok_share_7514fe.txt lines 6070-6090 (BB_C_Equations items 1310-1350)
- grok_share_7514fe.txt lines 6300-6500 (full reionization/BBN/CMB catalogue)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological)
- PAPER_200: Um Universal Magnetism Catalogue
- Planck 2018 Collaboration: CMB and reionization parameters